Department of Civil Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: I Semester B.E. CIVIL
Course Code & Title: GE3151 Problem solving and python programming
Name of the Faculty member (s): Ms.M.Preethi Ram

Innovative Practice Description

Unit / Topic: Unit I / Pseudo Code

Course Outcome: CO1

Topic Learning Outcome: TLO3

Activity Chosen: Mind Map

Justification:

Pseudo code in Python have various rules and sub topics. A mind map enables students to organize this information in a hierarchical and structured manner, making it easier to understand the relationships between different elements. Drawing a mind map is an active learning activity that engages students in the process of synthesizing and summarizing information. It encourages them to think critically about the topic and actively participate in their own learning

- **Time Allotted for the Activity:** 25 minutes

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 30 minutes. Instructor gave a program to the students.
- Based on the discussion and the teacher asked the students to draw a mind map related to the topic within 25 minutes.
- Each student created a mind map on Pseudo code.
- Students represent the concept taught in the class by visual representation.
- Each student has their unique way of flowchart based on their understanding about the topic of Pseudo code.
- Students can able to recollect the points taught in the class and able to visual it in an best manner.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PSO3
CO1	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO3
	2	1	1	1
Justification for correlation	Students can able to use the flow of control for for providing solution to the problems by using basics mathematics based on the Pseudo code.	Students can able to understand the sequence, Selection, formulated to solve the simple and complex engineering problems by using pseudocode.	Students can be able to provide solution to the real time decision making problems by using pseudo code.	Students can be able to apply the pseudo codes in python program to provide the solution for real time problems and coding related problems.

- **Images / Screenshot of the practice:**

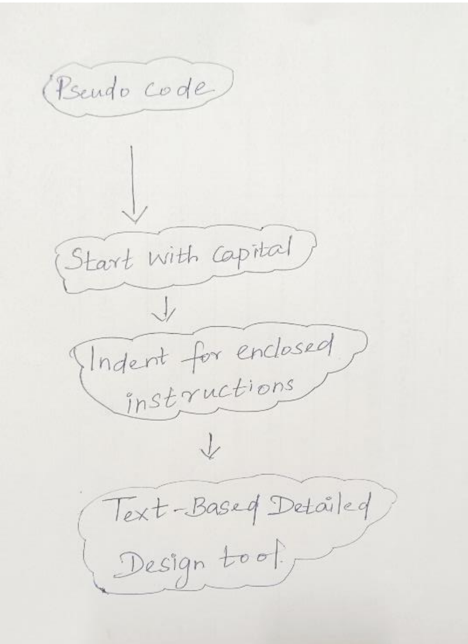

Fig 1: Sample Sheet of activity of the theStudent Dharun

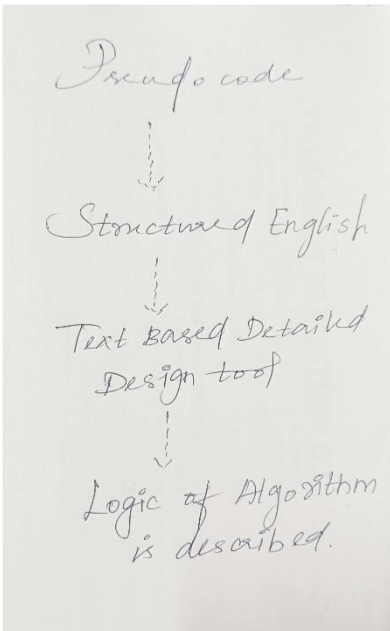

Fig 2: Sample Sheet of activity of Student Dhanusri

Reflective Critique:

Feedback of practice from students and other stakeholders:

- Students expressed that they understood the concept very well and the pictorial representation made them to understand the topic clearly.
- This helped the students in recalling the topic taught on that specific day, generating new ideas about the topic.

Benefit of the practice:

- From this activity, the students can get more clarity in the particular topic.
- Through this activity, the students are able to remember and describe the various

dictionary operations and it methods clearly

Challenges faced in implementation:

- Some of the students feel difficult to depict the concepts pictorially
- Some of them need some more clarification on the dictionary operations.
- Additional examples given to make them to understand the concept easily.

References:

https://www.ritrjpm.ac.in/images/computer-science/Mind%20Map.pdf
https://www.ritrjpm.ac.in/images/computer-science/43.CS6703_MindMap.pdf
https://www.ritrjpm.ac.in/images/computer-science/5_CS8591_Mindmap.pdf
https://www.lucidchart.com/pages/how-to-make-a-mind-map

Signature of Faculty Member **HOD**